

FoodTuck Website Testing Report

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Overview

This report provides a comprehensive review of the testing process, error handling, and backend improvements conducted for the FoodTuck Q-Commerce marketplace. The objective was to validate the platform's functionality, responsiveness, and reliability while adhering to industry-standard testing practices.

Test Cases and Results

1. Product Listing Page Validation

Test Case ID: TC001

- **Steps:** Navigate to the products page and verify the product display.
- **Expected Result:** Products should load correctly.
- **Actual Result:** Products displayed as expected.
- **Severity Level:** Low
- **Remarks:** No issues found.

2. API Error Handling

Test Case ID: TC002

- **Steps:** Disconnect API and refresh the page.
- **Expected Result:** An error message should appear.
- **Actual Result:** Error message displayed correctly.
- **Severity Level:** Low
- **Remarks:** Handled efficiently.

3. Cart Functionality

Test Case ID: TC003

- **Steps:** Add items to the cart and navigate to the cart page.
- **Expected Result:** Added products should appear in the cart.
- **Actual Result:** Products displayed correctly.
- **Severity Level:** Medium
- **Remarks:** Functions as expected.

4. Mobile Responsiveness

Test Case ID: TC004

- **Steps:** Test website using mobile device simulation.
- **Expected Result:** Content should adjust dynamically to screen size.
- **Actual Result:** Responsive design confirmed.
- **Severity Level:** Medium
- **Remarks:** No issues detected.

5. Search Bar Validation

Test Case ID: TC005

- **Steps:** Search for non-existent items.
- **Expected Result:** "No products found" message should be displayed.
- **Actual Result:** No message appeared.
- **Severity Level:** High
- **Remarks:** Requires immediate fixing.

6. Dynamic Product Detail Pages

Test Case ID: TC006

- **Steps:** Click on a product card and verify the correct detail page opens.
- **Expected Result:** Product details page should display the correct information.
- **Actual Result:** Works as intended.
- **Severity Level:** Low
- **Remarks:** No issues found.

7. Cross-Browser Compatibility

Test Case ID: TC007

- **Steps:** Open the website in Chrome, Firefox, and Edge.
- **Expected Result:** Consistent layout and functionality across browsers.
- **Actual Result:** No compatibility issues found.
- **Severity Level:** Low
- **Remarks:** Fully functional on all tested browsers.

Performance Optimization

Tools Used: Lighthouse, GTmetrix

Optimization Steps:

1. Compressed images via TinyPNG.
2. Implemented lazy loading for assets.
3. Minimized unused CSS and JavaScript.
4. Enabled browser caching for faster page loads.

Results:

- Initial page load time reduced to under 2 seconds.
 - Improved Lighthouse performance score: **95/100**.
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Error Handling

- **API Failures:** Managed via try-catch blocks.
- **Fallback UI:** Displayed user-friendly messages such as "No products available."

Example Implementation:

```
try {
  const data = await fetchProducts();
  setProducts(data);
} catch (error) {
  console.error("Failed to fetch products:", error);
  setError("Unable to load products. Please try again later.");
}
```

Cross-Browser and Device Testing

- **Browsers Tested:** Chrome, Firefox, Safari, Edge.
- **Devices Tested:** Desktop, tablet, iPhone 12.

Results:

- Fully responsive across all tested devices and browsers.
 - No layout inconsistencies detected.
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Security Measures

- **Input Validation:** Prevents injection attacks.
 - **Secure Communication:** HTTPS enabled.
 - **API Security:** Keys stored in environment variables.
 - **Tools Used:** OWASP ZAP, Burp Suite.
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Challenges and Resolutions

1. **Search bar failure:** The "No products found" message did not appear.
 - **Resolution:** Scheduled for fixing in the next sprint.
 2. **Performance bottlenecks due to large images.**
 - **Resolution:** Successfully optimized image sizes.
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Conclusion

The FoodTuck website successfully met most functional and non-functional testing criteria. The primary concern remains the search bar issue, which requires resolution before deployment. Overall, the platform is optimized for performance, responsiveness, and security, making it suitable for real-world use after the necessary fixes.

Next Steps

1. Fix search bar dropdown issue.
 2. Conduct additional User Acceptance Testing (UAT).
 3. Re-test post-fix implementation.
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Attachments

- CSV-based testing report.
- Lighthouse and GTmetrix audit logs.
- Security testing summary report.